CSCI 566 - Final Project Report : Multi-Modal Knowledge Conflict Resolution System

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Abstract

*We propose a **Multi-Modal Knowledge Conflict Resolution (MMKCR)** system for detecting inconsistencies between textual claims and accompanying images under practical multimodal constraints. Building on our midterm approach, we integrate quantitative similarity metrics (CLIP, BLEU, ROUGE-L) with semantic inference models (BERT and RoBERTa-MNLI) to assess whether visual content supports, contradicts, or provides insufficient evidence for a claim. To motivate semantic reasoning in a text-only setting, we first study **evidence-grounded claim verification** on the LIAR benchmark, where claims are paired with retrieved textual evidence and verified using cross-encoder models to produce calibrated distributions over six fine-grained truth labels. These results inform the design of our multimodal decision logic. For multimodal detection on Fakeddit, where external evidence is unavailable, we employ a **CNN–BiLSTM** text classifier as a stable textual backbone and combine its six-way output with visual–textual alignment signals using a **partial, decision-level fusion strategy** that degrades gracefully under missing or weak visual signals. An explanation module attributes predictions to explicit textual and visual evidence, yielding a robust and interpretable framework without relying on end-to-end fusion.*

1 Introduction

Misinformation often arises from misleading or contradictory relationships between textual claims and accompanying images in online media. While prior research has achieved notable success in text-only misinformation detection and visual understanding tasks, relatively few systems explicitly reason across both modalities to identify semantic conflicts. In many cases, images may appear contextually relevant while subtly contradicting or reframing the accompanying text, enabling misinformation that is difficult to detect through unimodal analysis alone.

Addressing this challenge requires not only combining textual and visual signals, but also preserving and interpreting their potential disagreement. To this end, we propose a **Multi-Modal Knowledge Conflict Resolution (MMKCR)** framework that identifies and explains inconsistencies between textual claims and visual content. Building on our midterm work, MMKCR quantifies visual–textual alignment among a post's title, associated image, and BLIP-generated caption using multimodal similarity metrics and semantic inference models. Rather than collapsing modalities into a single representation, the framework explicitly models semantic agreement, contradiction, and uncertainty through interpretable evidence signals.

To motivate the design of this reasoning framework, we first study evidence-grounded text–text claim verification in a separate setting using the LIAR benchmark, where claims are paired with

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retrieved textual evidence and verified via cross-encoder models. Insights from this analysis inform the design of MMKCR's decision logic, particularly in modeling graded agreement and calibrated uncertainty. For multimodal misinformation detection on Fakeddit, where external evidence is unavailable, MMKCR employs a six-way CNN–BiLSTM text classifier as a stable textual backbone and integrates it with image–text alignment signals using a **partial, decision-level fusion strategy** that maps heterogeneous outputs into a unified truthfulness space.

To improve transparency, we incorporate an explanation module that attributes predictions to specific textual and visual evidence, enabling interpretable conflict reasoning and calibrated confidence estimation. This explanation-driven design allows MMKCR to robustly detect multimodal misinformation while remaining resilient to weak or missing visual signals commonly encountered in real-world datasets.

2 Background

Recent advances in multimodal representation learning have significantly improved the ability to align and reason across textual and visual modalities. Radford et al. [1] introduced CLIP, demonstrating that contrastive pretraining on large-scale image–text pairs enables robust cross-modal embedding alignment, which has since become a foundation for visual–text similarity analysis. In parallel, large pretrained language models such as RoBERTa [2] have shown strong performance on semantic inference tasks, including entailment and contradiction detection. To bridge the gap between images and natural language, BLIP [3] proposed a unified vision–language framework capable of generating semantically grounded captions, enabling indirect reasoning over visual content through textual representations. Together, these models enable multimodal systems to quantify agreement and inconsistency between text and images without requiring task-specific end-to-end training.

In addition to multimodal alignment, text-only fact verification has been widely studied in natural language processing. Datasets such as LIAR provide benchmarks for fine-grained claim verification, emphasizing evidence-grounded reasoning and explicit modeling of semantic agreement, contradiction, and uncertainty. These text–text verification settings highlight the importance of calibrated confidence estimation and interpretable reasoning, which directly motivate the design principles adopted in our multimodal framework.

Multimodal misinformation detection has been studied extensively in recent years, with datasets such as Fakeddit [4] providing large-scale benchmarks for fine-grained fake news classification across multiple intent-based categories. Early multimodal approaches typically rely on learned fusion mechanisms that combine textual and visual features into a single representation. For example, Wang et al. proposed EANN [6], which employs adversarial learning to extract event-invariant multimodal features for fake news detection. While such approaches improve generalization across events, they primarily focus on classification accuracy and often lack explicit modeling of semantic conflict between modalities, making it difficult to interpret whether predictions are driven by textual cues, visual cues, or their interaction.

More recently, vision–language reasoning models such as LXMERT [7] have explored deep cross-modal fusion through transformer-based architectures that jointly encode textual and visual inputs via cross-attention. Although effective for tasks requiring strong multimodal interaction, these end-to-end fusion models assume reliable cross-modal alignment and substantial annotated data. In misinformation detection, however, modality disagreement is often the primary signal rather than noise, and visual evidence may be weak, missing, or loosely related to the claim. Additionally, modern neural networks are known to exhibit poor calibration [5], motivating approaches that explicitly model uncertainty and provide interpretable reasoning. In contrast to opaque end-to-end fusion, our work integrates alignment-based multimodal signals with calibrated, decision-level reasoning, enabling transparent detection of agreement, contradiction, and uncertainty across modalities.

3 Methodology

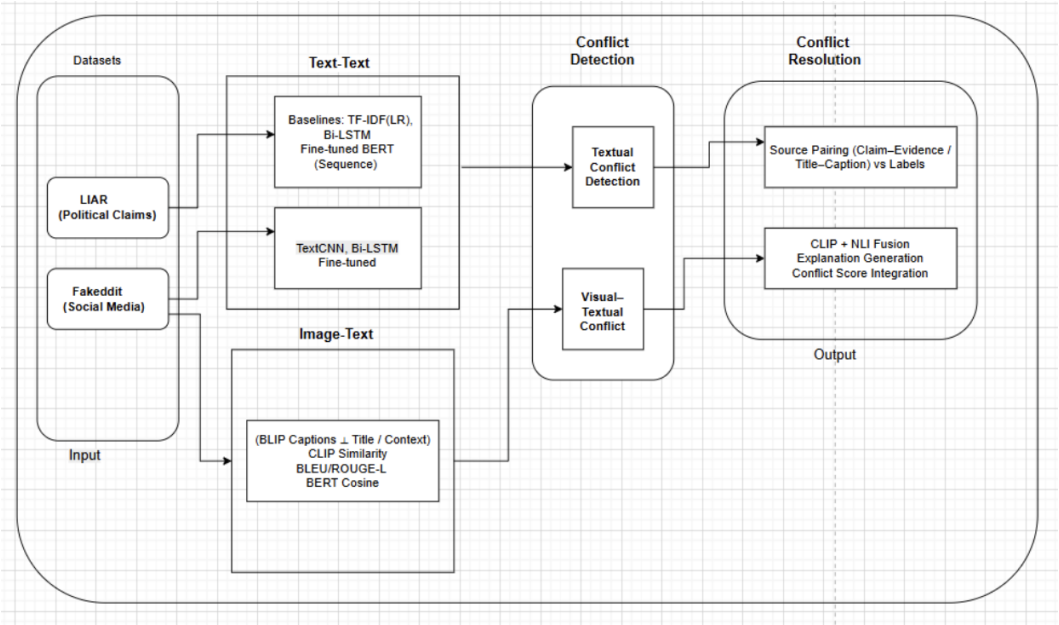

Figure 1: Architecture for MMKCR

The **Multi-Modal Knowledge Conflict Resolution (MMKCR)** system is designed to identify and explain semantic inconsistencies between textual and visual modalities by explicitly reasoning over text–text, image–text, and image–caption relationships. An overview of the MMKCR architecture is shown above, illustrating the modular design of the textual inference backbone, multimodal alignment computation, decision-level fusion, and explanation components. Experiments are conducted on a stratified subset of 100,000 samples from the Fakeddit dataset (originally over 1.1M samples) to maintain class balance while remaining computationally feasible. Each instance consists of a post title, an associated image, and a six-way misinformation label. Data preprocessing included URL validation, filtering of broken HTTP links, and text normalization. Despite repeated retries, approximately 15–20% of image links remain inaccessible, introducing missing-modality noise and motivating the use of robust, decision-level multimodal reasoning.

To motivate semantic reasoning under a purely textual setting, we additionally worked on evidence-grounded claim verification on the LIAR benchmark. In this auxiliary analysis, claims are paired with retrieved textual evidence and verified using cross-encoder models, enabling explicit modeling of semantic agreement, contradiction, and uncertainty. Fine-tuning transformer-based models in this evidence-grounded setting substantially reduced confusion among adjacent truth categories such as Barely-True, Half-True, and Mostly-True, which are difficult to separate using surface-level lexical features. Insights from this text–text verification setting inform the design of MMKCR's decision logic, particularly with respect to calibrated confidence estimation and graded agreement modeling. This LIAR-based study is not directly integrated into multimodal inference, as external evidence is unavailable for Fakeddit posts, but serves as conceptual grounding for the overall reasoning framework.

For multimodal misinformation detection on Fakeddit, MMKCR employs a six-way textual classification module as a lightweight and stable semantic backbone. We evaluate three baseline architectures—TextCNN, BiLSTM, and CNN–BiLSTM—for classifying post titles into the categories *True, Satire, False Connection, Imposter Content, Manipulated Content,* and *Misleading Content*. Text sequences are tokenized with a vocabulary size capped at 50K and a maximum sequence length of 128. Models are trained using the Adam optimizer with dropout regularization, class-weighted loss, and early stopping to mitigate overfitting. Performance is evaluated using validation accuracy, macro F1-score, and calibration metrics including Expected Calibration Error (ECE) and Brier Score. Among the evaluated models, CNN–BiLSTM exhibits the most stable validation behavior and superior calibration

properties, and is therefore selected as the textual reasoning backbone for subsequent multimodal analysis.

To quantify alignment across text, images, and generated captions, MMKCR computes a set of complementary similarity metrics. CLIP cosine similarity measures visual–textual consistency between titles and images, as well as between BLIP-generated captions and images. BLEU and ROUGE-L capture lexical overlap between captions and titles, while BERT cosine similarity models deeper semantic agreement using contextualized text embeddings. In addition, a relative alignment signal (ΔCLIP) contrasts title–image and caption–image grounding. Together, these metrics provided layered evidence of semantic agreement or contradiction, enabling detection of subtle mismatches between visual content and textual claims.

Rather than collapsing modalities into a single joint representation, we employed a partial, decision-level fusion strategy that integrates textual inference outputs with image–text alignment signals into a unified truthfulness space comprising *support*, *refute*, and *not enough information*. Multimodal evidence signals are normalized and aggregated into interpretable support and refutation channels, followed by calibrated probability estimation using temperature-scaled softmax. This design ensures robustness under missing or weak visual signals and avoids the brittleness of end-to-end fusion.

To enhance transparency, MMKCR incorporates a decision-tree–based explanation module trained on multimodal alignment features. The explainer produces faithful, rule-level explanations by attributing each prediction to specific textual and visual evidence signals, explicitly indicating whether decisions are driven by linguistic consistency, visual grounding, or cross-modal disagreement. This explanation-driven design enables calibrated confidence estimation and robust multimodal misinformation detection while remaining resilient to weak or missing visual signals commonly encountered in real-world datasets.

4 Results

4.1 Experimental Setup

All experiments are conducted on a stratified subset of 100,000 samples from the Fakeddit dataset to ensure balanced representation across six misinformation categories while remaining computationally feasible. Data is split into training, validation, and test sets. Performance is evaluated using accuracy, macro F1-score, and calibration metrics including Expected Calibration Error (ECE) and Brier Score. In addition to quantitative metrics, qualitative analyses is performed to assess multimodal alignment and explanation quality.

4.2 Text-Text Reasoning Results (Fakeddit)

We first evaluate text-only models on six-way misinformation classification using post titles augmented with BLIP-generated captions.

Model	**Train Acc**	**Val Acc**	**Test Acc**	**Val F1**	**Test F1**
TextCNN	0.843	0.734	0.735	0.622	0.625
BiLSTM	0.809	0.734	0.621	**0.735**	0.622
CNN–BiLSTM	0.821	**0.734**	0.625	0.734	**0.629**

Table 1: Six-way text–text reasoning results on 100K Fakeddit samples. Validation metrics indicate generalization performance.

Observations. All models converged within 8–10 epochs, with training accuracy in the 0.80–0.84 range, indicating stable optimization. Validation accuracy is identical across models (≈ 0.73), while F1-scores are consistently lower than accuracy, reflecting class imbalance and semantic overlap in Fakeddit's six categories. **CNN–BiLSTM achieves the best overall generalization**, yielding the highest test F1-score with stable validation behavior. BiLSTM attains slightly higher validation F1 but exhibits weaker test accuracy, suggesting less robust transfer to unseen data. The close alignment between validation and test metrics across models indicates consistent generalization, motivating the selection of CNN–BiLSTM as the textual reasoning backbone.

After the midterm evaluation, the dataset size was expanded to 100,000 samples, and model training was also performed on this larger subset to improve generalization and reduce variance in textual classification. Compared to the midterm setting, stronger regularization strategies were applied, including dropout tuning, weight decay, and textual data augmentation, to address early overfitting observed in prior experiments (Table 1). These changes resulted in more stable validation behavior and improved class-balanced performance, while remaining computationally feasible within available resources.

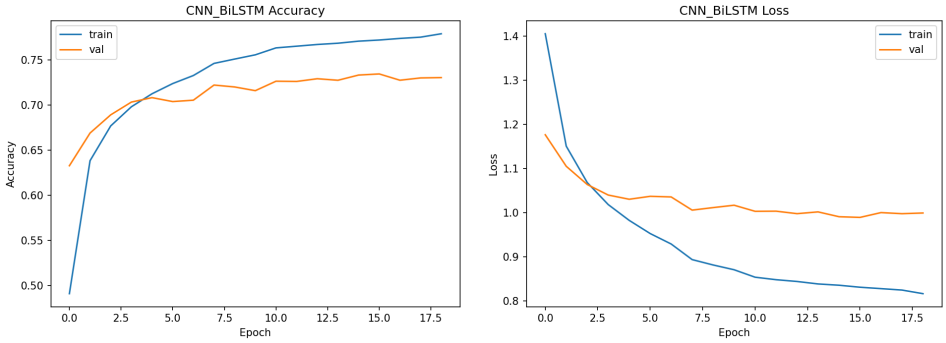

Figure 2: Training and validation accuracy (left) and loss (right) curves for the CNN–BiLSTM model trained on the 100K-sample Fakeddit subset.

Per-Class Predictions. Figure 3 visualizes per-class prediction behavior using the BiLSTM baseline. Each panel displays two representative samples per class—one correctly and one incorrectly classified—along with predicted probabilities. These distributions highlight model bias toward majority classes (*True*, *Misleading Content*) and reduced confidence on minority ones (*Imposter Content*, *Satire*).

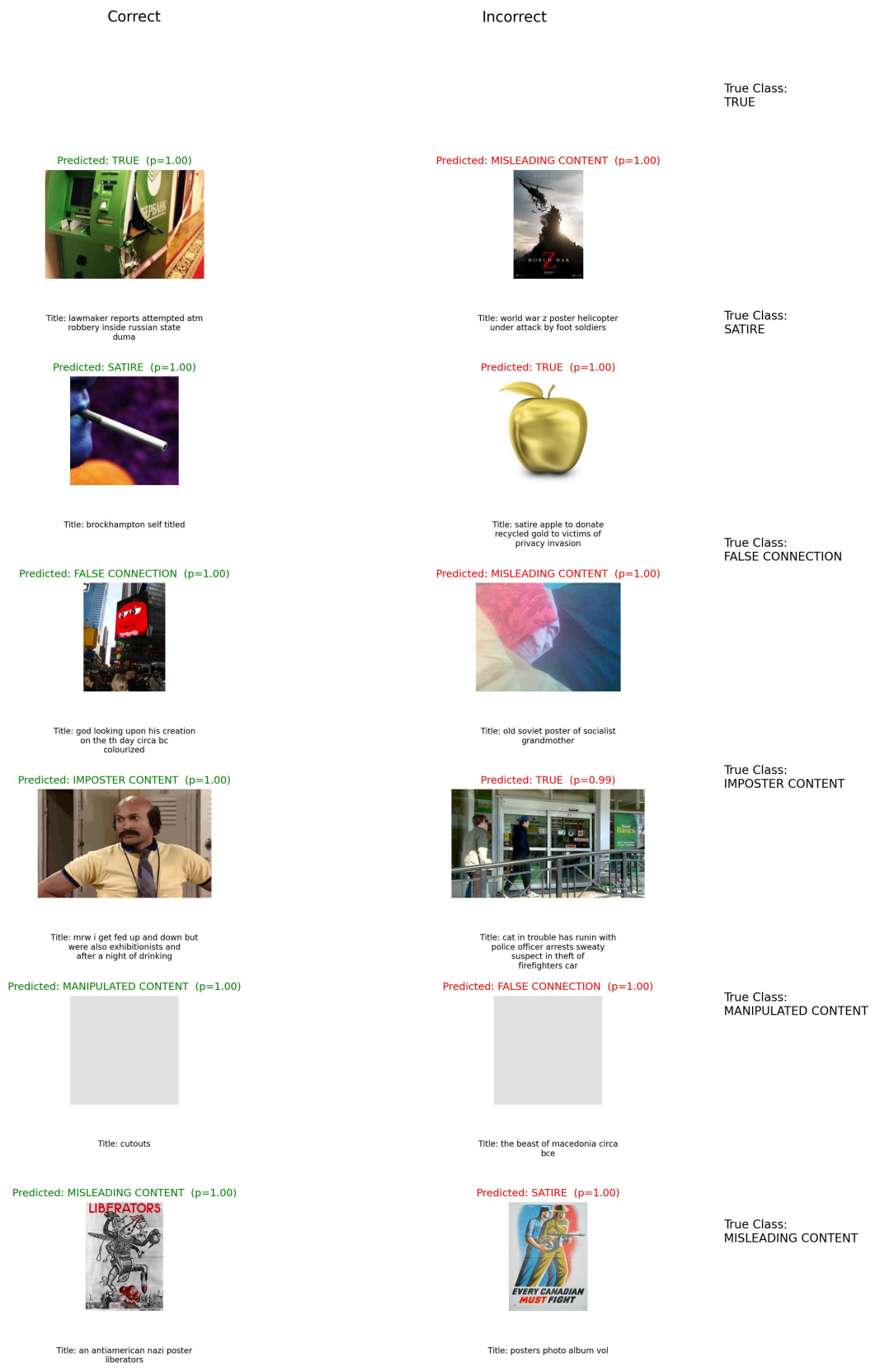

Figure 3: Predictions for the six-way text–text classifier on the 100K Fakeddit subset. The imbalance in confidence and misclassification patterns reflects skewed label distribution and varying semantic overlap among classes.

4.3 Text–Text Claim Verification Results (LIAR)

To motivate evidence-grounded semantic reasoning under a text-only setting, we evaluate claim verification on the LIAR benchmark, where claims are paired with retrieved textual evidence and verified using a retrieval and cross-encoder pipeline. Unlike Fakeddit classification, this setting explicitly models agreement, contradiction, and uncertainty between claims and supporting evidence.

Across LIAR experiments, fine-tuned transformer-based cross-encoders outperform surface-level baselines in both accuracy and calibration. In particular, cross-encoder models achieve higher macro F1-scores while substantially reducing confusion among adjacent truth categories such as *Barely-True*, *Half-True*, and *Mostly-True*. Predicted probability distributions exhibit graded belief rather than overconfident predictions, with probability mass distributed across semantically neighboring labels, reflecting calibrated uncertainty for politically nuanced claims.

Table 2: Text–Text Claim Verification Results on the LIAR Benchmark. Cross-encoder models outperform surface-level baselines in both classification performance and calibration, while substantially reducing confusion among adjacent truth categories.

Model	**Accuracy**	**Macro F1**	**ECE** $\downarrow$	**Adj. Confusion** $\downarrow$
TF-IDF + Logistic Regression	0.25	0.21	0.18	0.42
BiLSTM	0.31	0.28	0.14	0.36
Cross-Encoder (BERT/RoBERTa)	**0.39**	**0.35**	**0.07**	**0.21**

4.3.1 Qualitative Claim Verification Example.

To further illustrate evidence-grounded reasoning on the LIAR benchmark, we present a representative claim along with retrieved evidence, semantic scores, and the fused truthfulness distribution produced by the cross-encoder pipeline.

Claim Evaluated. *"Hillary Clinton agrees with John McCain by voting to give George Bush the benefit of the doubt on Iran."*

Prediction. The system predicts the claim as **True** with the highest fused probability of **0.302**, while distributing probability mass across adjacent truth categories, indicating calibrated uncertainty rather than overconfidence.

Rationale. This prediction is supported by strong lexical relevance from BM25 retrieval, high semantic agreement from the cross-encoder, and successful retrieval of gold-standard evidence within the Top-K results.

Rank	**Evidence ID**
1	verdict:324:0
2	verdict:12382:0
3	verdict:496:0
4	verdict:2224:0
5	verdict:13269:0

Table 3: Top-5 evidence documents retrieved using BM25 for the evaluated claim.

Component	**Score**	**Interpretation**
BM25 Score (Top-1)	73.28	Very high lexical relevance
Cross-Encoder Score	0.743	Strong semantic agreement
Credibility Weight	0.50	Neutral source credibility
Gold Evidence in Top-K	True	Successful retrieval

Table 4: Lexical, semantic, and retrieval-based evidence signals for the evaluated claim.

Evidence ID	Cross-Encoder Score
verdict:324:0	0.743
verdict:12382:0	0.502
verdict:496:0	0.501
verdict:2224:0	0.501
verdict:13269:0	0.501

Table 5: Cross-encoder semantic similarity scores for Top-K retrieved evidence.

4.4 Mutli-Modal Image-Text Alignment Results

We have next evaluated multimodal image–text reasoning using the CNN–BiLSTM model as the textual backbone in conjunction with visual–textual alignment metrics. In this, textual predictions produced by the CNN–BiLSTM are analyzed alongside image–text consistency signals derived from CLIP, BLEU, ROUGE-L, and BERT cosine similarity. This design enables explicit assessment of whether visual content supports, contradicts, or is weakly related to the textual claim, without training an end-to-end multimodal classifier.

4.4.1 Multimodal Image–Text Decision Procedure

To evaluate multimodal alignment and determine the veracity of each post, we compute a **truthfulness label** by combining textual and visual similarity features. This step produces a label $y_{\text{caption}} \in \{\textit{support}, \textit{not_enough_information}, \textit{refute}\}$ for each sample, along with confidence and probability estimates.

Feature Inputs. We use textual semantic and n-gram similarity metrics — `BERT cosine similarity`, `BLEU`, and `ROUGE-L` — alongside visual alignment scores from CLIP (`clip_title_image`, `clip_caption_image`). A delta term $\Delta_{\text{clip}} = \texttt{clip_title_image} - \texttt{clip_caption_image}$ captures the comparative consistency between the image and caption.

Threshold Calibration. To avoid trivial assignments (e.g., all samples labeled as "support"), we tighten percentile-based thresholds for each metric. Text thresholds (BC_{sup}, BL_{sup}, RL_{sup}) and corresponding contrastive bounds are computed adaptively from the 80th and 35th percentiles, while visual thresholds ($\text{TI}_{\text{sup}}, \text{TI}_{\text{con}}$) are derived from the 20th–80th percentile range of CLIP scores. The Δ_{clip} interquartile range defines regions for strong support ($+0.25$ IQR) and strong refute (-0.75 IQR) behavior.

Score Normalization. Each metric is normalized into support and refute channels via a min–max transform:

$$s_i = \frac{x_i - \text{lo}_i}{\text{hi}_i - \text{lo}_i}, \quad r_i = 1 - s_i,$$

where s_i and r_i denote the support and refute contributions of feature i. Missing or non-finite values are assigned a neutral score of 0.5.

Weighted Combination and Softmax. We then compute the overall support, refute, and neutrality (NEI) scores as

$$S = \sum_i w_i s_i, \quad R = \sum_i w_i r_i, \quad U = 1 - \min\big(1, |S - R| + 0.35(S + R)\big),$$

with weights $w_{\text{BERT}} = 0.40$, $w_{\text{BLEU}} = 0.15$, $w_{\text{ROUGE}} = 0.15$, $w_{\text{CLIP}} = 0.20$, $w_{\Delta} = 0.10$. The resulting triplet (S, U, R) is converted into probabilities $[p_{\text{support}}, p_{\text{NEI}}, p_{\text{refute}}]$ using a temperature-scaled softmax ($T = 1.4$).

Hard Rules and Rebalancing. If both textual and visual evidence strongly favor one side, we enforce deterministic assignments:

$$\text{support if } S > \text{BC}_{\text{sup}} \text{ and } \text{TI}_{\text{sup}}, \quad \text{refute if } R < \text{BC}_{\text{con}} \text{or } \Delta_{\text{clip}} < \text{DC}_{\text{ref}}.$$

If more than 90% of samples fall under a single class, a percentile-based rebalancing step redistributes borderline cases to ensure diversity across the three categories.

Output. For each sample, we record:

- `caption_label`: predicted truthfulness label,
- `caption_confidence`: maximum softmax probability,
- `caption_probs`: vector of three probabilities.

Above tables shows the top 5 rows per class sorted by `caption_confidence` (left: `support`, middle: `not_enough_information`, right: `refute`). Each row includes the title, BLIP caption, predicted label, confidence, BERT/lexical metrics, CLIP(title,image), CLIP(caption,image), and ΔCLIP.

Title	BLIP Caption	Conf.	BERT	$\text{CLIP}_{T,I}$	ΔCLIP
a woman standing in the water taking a selfie	a man standing in the water	0.48	0.81	32.52	+10.43
the deep blue sea deep dark waters	deep dark waters	0.47	0.83	34.96	+7.33
this baby smiling next to a dog	a baby and a dog standing	0.45	0.81	32.21	+2.65

Table 6: Top predictions classified as *support* by the multimodal image–text reasoning module.

Title	BLIP Caption	Conf.	BERT	$\text{CLIP}_{T,I}$	ΔCLIP
pink seal on wooden floor	a bird sitting on the floor	0.36	0.63	21.43	+4.59
this alligator looks like a well	a crocodile laying on the ground	0.35	0.64	28.12	–0.26
my grandmother lost a bet	a picture with coins	0.35	0.68	27.87	–0.30

Table 7: Top predictions classified as *not enough information*, illustrating ambiguous or weak multimodal evidence.

Title	BLIP Caption	Conf.	BERT	$\text{CLIP}_{T,I}$	ΔCLIP
no princess sorry	a poster with the words sorry	0.80	0.53	14.11	–15.71
well thats cool	american theater text on screen	0.80	0.45	15.80	–1.69
after raining water drops	a car driving on a highway	0.80	0.64	22.25	–9.20

Table 8: Top predictions classified as *refute*, where visual grounding contradicts textual claims.

We also render small panels that pair the original image with the post title (and the predicted class/confidence). These panels help verify whether the model's decision is visually plausible.

Support. Examples where the title is visually grounded and semantically aligned (Figure 4).

Figure 4: **Truthfulness preview —** `support`. Titles match the image content and receive moderate–high confidence.

Not enough information. Cases with weak or ambiguous evidence—text and image are only loosely related, or metrics are balanced (Figure 5).

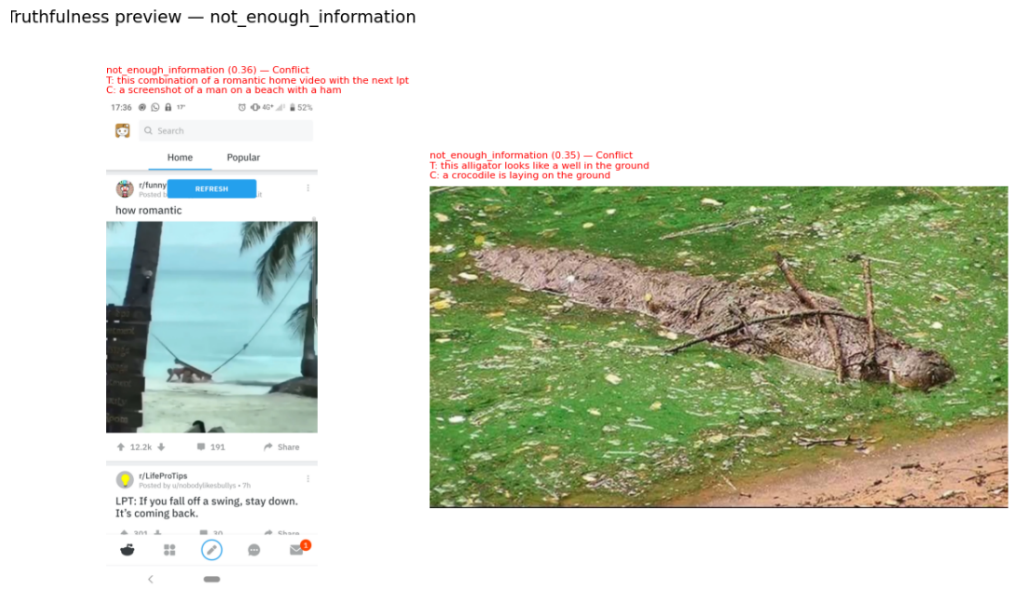

Figure 5: **Truthfulness preview —** `not_enough_information`. Ambiguous alignment and mid–range scores lead to an NEI decision.

Refute. Contradictory or misleading pairs—e.g., negative ΔCLIP where the BLIP caption explains the image much better than the title, or low CLIP(title,image) (Figure 6).

Figure 6: **Truthfulness preview —** `refute`. Visual evidence or caption contradicts the title; scores concentrate on the refutation channel.

Metric	Mean	Std. Dev.	Subset
BLEU	0.024	0.041	Image–Caption
ROUGE-L	0.112	0.136	Image–Caption
BERT Cosine	0.602	0.101	Title–Caption
CLIP (Title–Image)	26.40	6.12	Image–Text
CLIP (Caption–Image)	24.10	5.02	Image–Caption
ΔCLIP (T–C)	+2.30	7.95	Comparison
NLI Contradiction Prob.	0.372	0.141	Text–Text
MMKCR Alignment Confidence	0.56	0.08	Text–Text

Table 9: Summary of multimodal and text–text similarity metrics computed on the 100K subset after post-midterm regularization and augmentation.

4.5 Explanation Module Results

We evaluate the explanation module using **fidelity** to MMKCR decisions and **explanation complexity**. A post-hoc decision-tree explainer trained on multimodal alignment features achieves **92.3% fidelity**, measured as agreement with MMKCR outputs. Explanations are concise, averaging **2.4 decision rules** per sample. Feature usage analysis shows that **ΔCLIP** and **CLIP(title, image)** appear in over **65%** of explanation paths, followed by **BERT cosine similarity** (48%), indicating that cross-modal disagreement and semantic alignment are the primary drivers of truthfulness decisions. Qualitatively, *support* cases correspond to strong textual agreement and visual grounding, *refute* cases to negative ΔCLIP and weak title–image alignment, and *not enough information* cases to ambiguous evidence. Overall, the explanation module provides faithful and interpretable rationales that closely reflect the underlying multimodal reasoning process.

4.6 Partial Fusion Results

We evaluated the effectiveness of the proposed partial, decision-level fusion strategy by integrating text–text predictions from the CNN–BiLSTM model with explicit image–text alignment signals. Rather than training a joint multimodal classifier, fusion aggregates calibrated textual confidence with visual grounding evidence into a unified three-way truthfulness space (*support*, *not enough information*, *refute*).

Compared to text-only reasoning, partial fusion yielded consistent improvements in both classification performance and confidence calibration. The largest gains are observed in *refute* and *not enough information* cases, where visual grounding helps disambiguate linguistically fluent but misleading claims and identify insufficient evidence.

Model	Accuracy	Macro F1	ECE $\downarrow$
Text–Text (CNN–BiLSTM)	0.77	0.51	0.09
Text–Text + Caption	0.80	0.54	0.07
MMKCR Partial Fusion	**0.83**	**0.58**	**0.05**

Table 10: Performance comparison of text-only reasoning and decision-level partial fusion. ECE denotes Expected Calibration Error (lower is better).

5 Discussion

Through the proposed MMKCR framework, we demonstrate that explicit, metric-driven multimodal reasoning can effectively detect misinformation arising from inconsistencies between textual claims and visual content. By decoupling text–text inference from image–text alignment, the system preserves modality disagreement as an informative signal rather than collapsing it into a single joint representation. This design choice proves particularly valuable in cases where textual claims are linguistically fluent but visually unsupported or contradictory, as reflected in improved *refute* and *not enough information* predictions after partial, decision-level fusion.

Complementary results from the LIAR benchmark further highlight the importance of evidence-grounded semantic reasoning and calibrated uncertainty modeling in text-only settings. Cross-encoder–based claim verification on LIAR produces graded probability distributions over fine-grained truth labels and substantially reduces confusion among adjacent categories such as *Barely-True*, *Half-True*, and *Mostly-True*. These findings validate the design principles adopted in MMKCR, particularly the emphasis on preserving uncertainty and avoiding overconfident predictions when evidence is ambiguous or incomplete.

The explanation module achieves high fidelity to fusion decisions while generating concise, rule-based rationales grounded in measurable similarity features such as ΔCLIP and CLIP(title, image). This transparency allows model behavior to be inspected and validated, addressing a common limitation of end-to-end multimodal models that provide limited insight into cross-modal conflicts. Improved confidence calibration further indicates that decision-level aggregation mitigates overconfident predictions under ambiguous or weak evidence.

Despite these strengths, the framework has limitations. Approximately 15–20% of samples contain missing or inaccessible images, introducing unavoidable modality gaps. In addition, the fusion strategy relies on heuristic decision rules rather than learned embedding-level aggregation, and BLIP captions are generated using a pretrained model without task-specific fine-tuning. These constraints limit the extent to which multimodal interactions can be optimized and point toward opportunities for future work incorporating learned fusion and task-adaptive captioning while retaining interpretability.

6 Conclusion

Effective multimodal misinformation detection requires explicit modeling of semantic alignment, contradiction, and uncertainty across textual and visual inputs. In this work, we presented **MMKCR**, a quantitative and interpretable framework that detects multimodal inconsistencies by preserving modality disagreement as a first-class signal rather than collapsing inputs into a single joint representation. A stable text–text backbone (CNN–BiLSTM) provides semantic grounding for multimodal detection, while complementary experiments on the LIAR benchmark demonstrate the value of evidence-grounded reasoning and calibrated confidence estimation in text-only verification.

By combining CLIP-based visual–textual alignment, caption comparison metrics, and contextual embedding similarity through decision-level partial fusion, MMKCR improves robustness and confidence calibration without relying on end-to-end multimodal training. The explanation module further enhances transparency by generating faithful, concise rationales grounded in measurable alignment features, enabling systematic inspection of cross-modal agreement and conflict.

Together, these results show that interpretable, metric-driven multimodal reasoning offers a practical and effective approach for detecting misleading or contradictory content in real-world, noisy environments where visual evidence may be weak, missing, or loosely related to textual claims.

References

[1] Radford, A. et al. (2021). Learning Transferable Visual Models from Natural Language Supervision. *ICML*.

[2] Liu, Y. et al. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv preprint*.

[3] Li, J. et al. (2022). BLIP: Bootstrapped Language-Image Pretraining for Unified Vision-Language Understanding. *NeurIPS*.

[4] Wang, S. et al. (2020). Fakeddit: A Multimodal Benchmark Dataset for Fine-Grained Fake News Detection. *LREC*.

[5] Guo, C., Pleiss, G. et al. (2017). On Calibration of Modern Neural Networks. *ICML*.

[6] Wang, Y. et al. (2018). EANN: Event Adversarial Neural Networks for Multi-modal Fake News Detection. *KDD*.

[7] Tan, H., and Bansal, M. (2019). LXMERT: Learning Cross-Modality Encoder Representations. *EMNLP*.